Question 1 (4 marks)

Silicon has a diamond cubic crystal structure with a lattice constant of 5.43 A. Find the number density of silicon atoms in atoms/cm³.

$$N = \frac{No.\ of\ atoms\ per\ unit\ cell}{Unit\ cell\ volume}$$

[4 marks]

Question 2 (6 marks)

A silicon crystal is doped with 10^{16} /cm^3 arsenic atoms (Group V). Find the carrier concentration (holes and electrons) and the Fermi level at room temperature (300K). Draw a band diagram showing the position of the Fermi level ($n_i = 10^{10}$ /cm^3).

[6 marks]

Question 3 (15 marks)

Consider a slab of phosphorus-doped Silicon (Si) that is at room temperature and in thermal equilibrium. Silicon is a Group IV element, while phosphorus is a Group V element. The phosphorus has a concentration of 3.0×10^{15} cm^{-3} ($n_i = 10^{10}$ /cm^3).

a. If an electric field of E = 100 V/cm is applied across the silicon slab, calculate the electron drift current density.

[3 marks]

b. Under the same electric field, compute the hole drift current density.

[3 marks]

c. If the electron concentration varies linearly from 3.0×10^{15} cm^{-3} at x = 0 to 2.0×10^{15} cm^{-3} at x = 10 μm, calculate the electron diffusion current density. Use $D_n = 35$ cm^2/s.

[4 marks]

d. For an applied voltage $V_a = 3$ V, a current $I = 1.0$ mA is required. Design suitable values of cross-sectional area A and length L so that the current density is kept at 20 A/cm^2.

[5 marks]

Question 4 (25 marks)

A Si PN junction with a cross-sectional area of 10^{-4} cm^2 has the following properties at T = 300 K.

p region	*n region*
$N_A = 10^{17}$ cm^{-3}	$N_D = 5 \times 10^{15}$ cm^{-3}
$\tau_{po} = 10^{-7}$ s	$\tau_{no} = 10^{-6}$ s
$\mu_n = 750$ cm^2/V-s	$\mu_n = 1150$ cm^2/V-s
$\mu_p = 300$ cm^2/V-s	$\mu_p = 420$ cm^2/V-s

a. Calculate the built-in potential of the PN junction. Use intrinsic carrier concentration of Si, $n_i = 1.0 \times 10^{10}$ cm^{-3}.

[3 marks]

b. Determine the width of the depletion region and the depletion width in each side of the junction. Relative permittivity for Si, $\varepsilon_r = 11.7$.

[6 marks]

c. Draw the energy band diagram of a silicon PN junction at equilibrium. Clearly label conduction band, E_C, valence band, E_V, Fermi level, E_F, Built-in potential (V_{bi}), depletion region boundaries.

[4 marks]

d. Compute the reverse saturation current. Assume $D_n = 35$ cm^2/s and $D_p = 12$ cm^2/s.

[3 marks]

e. Compare the PN junction diode, Schottky diode and a heterostructure in terms of their structure and materials used. Draw a simple labelled diagram for each device and state one application for each.

[9 marks]

- END OF PAPER -

APPENDIX

Physical Constants:

Boltzmann's constant	$k = 1.38 \times 10^{-23}$ J/K $k = 8.62 \times 10^{-5}$ eV/K
Electronic charge	$q = 1.60 \times 10^{-19}$ C
Free electron rest mass	$m_o = 9.11 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_o = 8.85 \times 10^{-12}$ F/cm $= 8.85 \times 10^{-14}$ F/cm
Plank's constant	$h = 6.625 \times 10^{-34}$ J-s $= 4.135 \times 10^{-15}$ eV-s $\hbar = 1.054 \times 10^{-34}$ J-s
Speed of light in vacuum	$c = 2.998 \times 10^{10}$ cm/s
Thermal voltage (T = 300 K)	$V_t = kT/q = 0.0259$ V $kT = 0.0259$ eV

Silicon, Gallium Arsenide, and Germanium Properties (T = 300 K):

Property	Si	GaAs	Ge
Lattice constant (Å)	5.43	5.65	5.65
Bandgap energy (eV)	1.12	1.42	0.66
Electron affinity, χ (V)	4.01	4.07	4.13
Effective density of states in conduction band, N_c (cm^{-3})	2.8×10^{19}	4.7×10^{17}	1.04×10^{19}
Effective density of states in valence band, N_v (cm^{-3})	1.04×10^{19}	7.0×10^{18}	6.0×10^{18}
Electron Mobility, μ_n (cm^2/V-s)	1350	8500	3900
Hole Mobility, μ_p (cm^2/V-s)	480	400	1900
Intrinsic Carrier Concentration, n_i (cm^{-3})	1.5×10^{10}	1.8×10^{6}	2.4×10^{13}

Carrier Mobility and Diffusion Coefficient as a Function of Doping Concentration

